AWS Well-Architected Tool

# AWS Well-Architected Tool FTR_WAFR_PASSING - AWS Well-Architected Framework - PASSING_1 Report

AWS Account ID: 156345700632

# AWS Well-Architected Tool Report



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# Milestone properties

**Milestone name**
PASSING_1

**Date saved**
Nov 3, 2025 5:44 PM UTC

**Workload name**
FTR_WAFR_PASSING

**ARN**
arn:aws:wellarchitected:us-east-1:156345700632:workload/5cc3a1e4d8f3ea4e5cb22127eee57a1c

**Description**
sample of a passing FTR+WAFR

**Review owner**
derekvi

**Industry type**
-

**Industry**
-

**Environment**
Production

**AWS Regions**
US East (N. Virginia)

**Non-AWS regions**
-

## Account IDs

-

## Architectural design

-

## Application

-

  

# Lens overview

**Questions answered**

55/57

**Version**

AWS Well-Architected Framework, 25th Feb 2025

| Pillar | Questions answered |
| --- | --- |
| Operational Excellence | 11/11 |
| Security | 9/11 |
| Reliability | 13/13 |
| Performance Efficiency | 5/5 |
| Cost Optimization | 11/11 |
| Sustainability | 6/6 |

**Lens notes**

-

# Improvement plan

**Improvement item summary**

High risk: 8

Medium risk: 2

| Pillar | High risk | Medium risk |
|---|---|---|
| Operational Excellence | 1 | 0 |
| Security | 4 | 0 |
| Reliability | 3 | 2 |
| Performance Efficiency | 0 | 0 |
| Cost Optimization | 0 | 0 |
| Sustainability | 0 | 0 |

## High risk

### Operational Excellence

- OPS 11.How do you evolve operations?

### Security

- SEC 6.How do you protect your compute resources?
- SEC 7.How do you classify your data?
- SEC 8.How do you protect your data at rest?
- SEC 10.How do you anticipate, respond to, and recover from incidents?

| Reliability |
| --- |
| • REL 13.How do you plan for disaster recovery (DR)?<br>• REL 3.How do you design your workload service architecture?<br>• REL 1.How do you manage service quotas and constraints? |

| Performance Efficiency |
| --- |
| No improvements identified |

| Cost Optimization |
| --- |
| No improvements identified |

| Sustainability |
| --- |
| No improvements identified |

# Medium risk

| Operational Excellence |
| --- |
| No improvements identified |

| Security |
| --- |
| No improvements identified |

| Reliability |
| --- |
| • REL 9.How do you back up data?<br>• REL 12.How do you test reliability? |

| Performance Efficiency |
| --- |
| No improvements identified |

| Cost Optimization |
| --- |
| No improvements identified |

| Sustainability |
| --- |
| No improvements identified |

# Lens details

## Operational Excellence

**Questions answered**

11/11

**Question status**

- High risk: 1
- Medium risk: 0
- No improvements identified: 10
- Not Applicable: 0
- Unanswered: 0

**Pillar notes**

-

## 1. How do you determine what your priorities are?

No improvements identified

*This question has best practices marked as not applicable by the reviewer

### Selected choice(s)

-

### Not selected choice(s)

- None of these

### Best Practices marked as Not Applicable

- Evaluate compliance requirements

  Out of Scope

- Evaluate threat landscape

  Out of Scope

- Evaluate tradeoffs while managing benefits and risks

  Out of Scope

- Evaluate external customer needs

  Out of Scope

- Evaluate governance requirements

  Out of Scope

- Evaluate internal customer needs

  Out of Scope

### Notes

OPS_1

### Improvement plan

| 1. How do you determine what your priorities are? |
| --- |
| No risk detected for this question. No action needed. |

## 2. How do you structure your organization to support your business outcomes?

No improvements identified

*This question has best practices marked as not applicable by the reviewer

**Selected choice(s)**

- Mechanisms exist to manage responsibilities and ownership

**Not selected choice(s)**

- None of these

**Best Practices marked as Not Applicable**

- Operations activities have identified owners responsible for their performance

  Out of Scope

- Responsibilities between teams are predefined or negotiated

  Out of Scope

- Processes and procedures have identified owners

  Out of Scope

- Resources have identified owners

  Out of Scope

- Mechanisms exist to request additions, changes, and exceptions

  Out of Scope

**Notes**

OPS_2: Provide answers for the following:
1) Have you reviewed the AWS Shared Responsibility Models for Security and Resiliency (Yes/No)?: Yes

| 2. How do you structure your organization to support your business outcomes? |
| --- |
| 2) Do you have a mechanism for communicating resiliency responsibility to customers (Yes/No)?: Yes<br><br>**Improvement plan**<br><br>No risk detected for this question. No action needed. |

## 3. How does your organizational culture support your business outcomes?

No improvements identified

*This question has best practices marked as not applicable by the reviewer

**Selected choice(s)**

-

**Not selected choice(s)**

- None of these

**Best Practices marked as Not Applicable**

- Communications are timely, clear, and actionable

  Out of Scope

- Provide executive sponsorship

  Out of Scope

- Team members are empowered to take action when outcomes are at risk

  Out of Scope

- Escalation is encouraged

  Out of Scope

- Experimentation is encouraged

  Out of Scope

- Team members are encouraged to maintain and grow their skill sets

  Out of Scope

- Resource teams appropriately

  Out of Scope

**Notes**

| 3. How does your organizational culture support your business outcomes? |
| --- |
| OPS_3<br><br>**Improvement plan**<br><br>No risk detected for this question. No action needed. |

## 4. How do you implement observability in your workload?

No improvements identified

*This question has best practices marked as not applicable by the reviewer

### Selected choice(s)

-

### Not selected choice(s)

- None of these

### Best Practices marked as Not Applicable

- Implement application telemetry

  Out of Scope

- Implement user experience telemetry

  Out of Scope

- Implement dependency telemetry

  Out of Scope

- Implement distributed tracing

  Out of Scope

- Identify key performance indicators

  Out of Scope

### Notes

OPS_4

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### Improvement plan

No risk detected for this question. No action needed.

## 5. How do you reduce defects, ease remediation, and improve flow into production?

No improvements identified

*This question has best practices marked as not applicable by the reviewer

**Selected choice(s)**

-

**Not selected choice(s)**

- None of these

**Best Practices marked as Not Applicable**

- Fully automate integration and deployment
  Out of Scope
- Use build and deployment management systems
  Out of Scope
- Implement practices to improve code quality
  Out of Scope
- Use configuration management systems
  Out of Scope
- Make frequent, small, reversible changes
  Out of Scope
- Use multiple environments
  Out of Scope
- Perform patch management
  Out of Scope
- Share design standards

| 5. How do you reduce defects, ease remediation, and improve flow into production? |
| --- |
| Out of Scope<br>• Test and validate changes<br>Out of Scope<br>• Use version control<br>Out of Scope<br>**Notes**<br>OPS_5<br><br>**Improvement plan**<br>No risk detected for this question. No action needed. |

## 6. How do you mitigate deployment risks?

No improvements identified

*This question has best practices marked as not applicable by the reviewer

**Selected choice(s)**

-

**Not selected choice(s)**

- None of these

**Best Practices marked as Not Applicable**

- Automate testing and rollback

  Out of Scope

- Employ safe deployment strategies

  Out of Scope

- Plan for unsuccessful changes

  Out of Scope

- Test deployments

  Out of Scope

**Notes**

OPS_6

**Improvement plan**

No risk detected for this question. No action needed.

## 7. How do you know that you are ready to support a workload?

No improvements identified

*This question has best practices marked as not applicable by the reviewer

### Selected choice(s)

- Create support plans for production workloads

### Not selected choice(s)

- None of these

### Best Practices marked as Not Applicable

- Ensure a consistent review of operational readiness

  Out of Scope

- Make informed decisions to deploy systems and changes

  Out of Scope

- Ensure personnel capability

  Out of Scope

- Use playbooks to investigate issues

  Out of Scope

- Use runbooks to perform procedures

  Out of Scope

### Notes

OPS_7: Provide answers for the following:
1) Is your support tier Business of higher on all production AWS Accounts (Yes/No)?: Yes

### Improvement plan

| 7. How do you know that you are ready to support a workload? |
| --- |
| No risk detected for this question. No action needed. |

  

## 8. How do you utilize workload observability in your organization?

No improvements identified

*This question has best practices marked as not applicable by the reviewer

### Selected choice(s)

-

### Not selected choice(s)

- None of these

### Best Practices marked as Not Applicable

- Analyze workload logs
  Out of Scope
- Analyze workload metrics
  Out of Scope
- Analyze workload traces
  Out of Scope
- Create actionable alerts
  Out of Scope
- Create dashboards
  Out of Scope

### Notes

OPS_8

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### Improvement plan

No risk detected for this question. No action needed.

## 9. How do you understand the health of your operations?

No improvements identified

*This question has best practices marked as not applicable by the reviewer

**Selected choice(s)**

-

**Not selected choice(s)**

- None of these

**Best Practices marked as Not Applicable**

- Communicate status and trends to ensure visibility into operation
  Out of Scope
- Measure operations goals and KPIs with metrics
  Out of Scope
- Review operations metrics and prioritize improvement
  Out of Scope

**Notes**

OPS_9

**Improvement plan**

No risk detected for this question. No action needed.

## 10. How do you manage workload and operations events?

No improvements identified

*This question has best practices marked as not applicable by the reviewer

### Selected choice(s)

- Define a customer communication plan for service-impacting events
- Communicate status through dashboards

### Not selected choice(s)

- None of these

### Best Practices marked as Not Applicable

- Automate responses to events

  Out of Scope

- Define escalation paths

  Out of Scope

- Use a process for event, incident, and problem management

  Out of Scope

- Prioritize operational events based on business impact

  Out of Scope

- Have a process per alert

  Out of Scope

### Notes

OPS_10

### Improvement plan

No risk detected for this question. No action needed.

## 11. How do you evolve operations?

High risk

*This question has best practices marked as not applicable by the reviewer

### Selected choice(s)

-

### Not selected choice(s)

- Have a process for continuous improvement
- None of these

### Best Practices marked as Not Applicable

- Allocate time to make improvements

  Out of Scope

- Define drivers for improvement

  Out of Scope

- Implement feedback loops

  Out of Scope

- Perform knowledge management

  Out of Scope

- Perform operations metrics reviews

  Out of Scope

- Perform post-incident analysis

  Out of Scope

- Document and share lessons learned

  Out of Scope

- Validate insights

## 11. How do you evolve operations?

Out of Scope

### Notes

OPS_11: Document how you implement the following best practice:
1) Do you do periodic architecture reviews (Yes/No)?: Yes
2) Have you done an architecture review in the last year (Yes/No)?: Yes
-- Note that if you are using this initial FTR as a starting point, you can create a milestone for the workload and then continue by completing all questions in the future after you have submitted for FTR.

### Improvement plan

- Have a process for continuous improvement

Ask an expert

# Security

### Questions answered

9/11

### Question status

- High risk: 4
- Medium risk: 0
- No improvements identified: 5
- Not Applicable: 0
- Unanswered: 2

### Pillar notes

-

## 1. How do you securely operate your workload?

No improvements identified

*This question has best practices marked as not applicable by the reviewer

### Selected choice(s)

- Secure account root user and properties
- Identify and validate control objectives

### Not selected choice(s)

- None of these

### Best Practices marked as Not Applicable

- Automate deployment of standard security controls

  Out of Scope
- Evaluate and implement new security services and features regularly

  Out of Scope
- Separate workloads using accounts

  Out of Scope
- Reduce security management scope

  Out of Scope
- Identify and prioritize risks using a threat model

  Out of Scope
- Stay up to date with security threats and recommendations

  Out of Scope

### Notes

SEC_1: Provide answers for the following:
1) Is the AWS Root Account used for any other tasks than the ones

| 1. How do you securely operate your workload? |
| --- |
| listed here: https://docs.aws.amazon.com/accounts/latest/reference/root-user-tasks.html (Yes/No)?: No<br>2) Do you have access keys provisioned for the Root User (Yes/No)?: No<br>3) Have you included your CIS Benchmark report as an artifact (Yes/No)?: Yes<br>**Improvement plan**<br>No risk detected for this question. No action needed. |

## 2. How do you manage identities for people and machines?

No improvements identified

*This question has best practices marked as not applicable by the reviewer

### Selected choice(s)

- Use strong sign-in mechanisms
- Use temporary credentials
- Store and use secrets securely
- Rely on a centralized identity provider
- Audit and rotate credentials periodically

### Not selected choice(s)

- None of these

### Best Practices marked as Not Applicable

- Employ user groups and attributes

  Out of Scope

### Notes

SEC_2: Provide answers for the following:
1) Is MFA enabled for all human users to access AWS Accounts -- this may be enabled in your IdP to federate access (Yes/No)?: Yes
2) Are the credentials and access keys rotated at most every 90 days (Yes/No)?: Yes
3) Do you have a list of all static keys and where they are used (Yes/No)?: Yes
4) Have you implemented monitoring of AWS CloudTrail logs to detect anomalous activity or other potential misuse (Yes/No)?: Yes
5) Do you have a runbook or SOP for revoking credentials in the event

## 2. How do you manage identities for people and machines?

you detect a misuse (Yes/No)?: Yes
6) Is a strong password policy enforced for users (Yes/No)?: Yes
7) Does every individual user have their own unique credentials which are not shared with other users (Yes/No)?: Yes
8) Was there an audit conducted for IAM users, roles and permissions in the last quarter (Yes/No)?: Yes
9) Are there credentials hard-coded in any application code (Yes/No)?: No
10) Do you use secret management tools for encryption, access control, and access logging (Yes/No)?: Yes
11) Do you encrypt end-user authentication credentials and passwords at rest (Yes/No)?: Yes
12) Do you use IAM roles for all AWS API access from AWS compute services (e.g. Amazon EC2 Instance, AWS Lambda Functions, etc) (Yes/No)?: Yes
13) For any cases where you must use static IAM credentials, are you storing them securely (Yes/No)?: Yes
14) Does your solutions architecture ensure provisioning read-only access to external IDs to customers (Yes/No)?: Yes
15) Does your solution architecture ensure that all external IDs for external AWS Accounts are unique (Yes/No)?: Yes
16) Does your solution architecture ensure any historical use of customer-provided IAM credentials are deprecated (Yes/No)?: Yes

**Improvement plan**

No risk detected for this question. No action needed.

## 3. How do you manage permissions for people and machines?

No improvements identified

*This question has best practices marked as not applicable by the reviewer

### Selected choice(s)

- Grant least privilege access
- Define permission guardrails for your organization
- Manage access based on lifecycle
- Reduce permissions continuously
- Share resources securely with a third party
- Analyze public and cross account access

### Not selected choice(s)

- None of these

### Best Practices marked as Not Applicable

- Define access requirements

  Out of Scope

- Establish emergency access process

  Out of Scope

- Share resources securely within your organization

  Out of Scope

### Notes

SEC_3: Provide answers for the following:
1) Are third parties given access to the production workload (Yes/No)?:
No
2) Do you provide temporary of static credentials (Yes/No)?:
3) Is the principal of least privilege followed when granting access to

| 3. How do you manage permissions for people and machines? |
| --- |
| users (Yes/No)?:<br>4) Is there a policy for when a user leaves the company or changes roles (Yes/No)?:<br>5) Is there a process in place to periodically review access levels of S3 buckets (Yes/No)?:<br>6) Do you use cross-account roles to access customer AWS Accounts (Yes/No)?:<br>7) Do you provide guidance for customers on how to setup their AWS Account(s) for external access by your AWS Accounts (Yes/No)?:<br>8) Is external ID being used to access another customer's AWS Account (Yes/No)?:<br>9) Do you generate the values for external IDs to provide to customers (Yes/No)?: |
| **Improvement plan**<br>No risk detected for this question. No action needed. |

  

## 4. How do you detect and investigate security events?

No improvements identified

*This question has best practices marked as not applicable by the reviewer

**Selected choice(s)**

-

**Not selected choice(s)**

- None of these

**Best Practices marked as Not Applicable**

- Configure service and application logging

  Out of Scope

- Capture logs, findings, and metrics in standardized locations

  Out of Scope

- Initiate remediation for non-compliant resources

  Out of Scope

- Correlate and enrich security events

  Out of Scope

**Notes**

SEC_4

---

**Improvement plan**

No risk detected for this question. No action needed.

## 5. How do you protect your network resources?

Unanswered

**Selected choice(s)**

-

**Not selected choice(s)**

- Create network layers
- Control traffic within your network layers
- Implement inspection-based protection
- Automate network protection
- None of these

**Best Practices marked as Not Applicable**

-

**Notes**

SEC_5: Provide answers for the following:
1) Does the default security group restrict all traffic (Yes/No)?:
2) Are all security groups configured to not all ingress from 0.0.0.0/0 to port 22 or 3389 (Yes/No)?:
3) Are all of the resources hosted in private subnets (Yes/No)?:

---

**Improvement plan**

Answer the question to view the improvement plan.

  

## 6. How do you protect your compute resources?

High risk

*This question has best practices marked as not applicable by the reviewer

### Selected choice(s)

-

### Not selected choice(s)

- Perform vulnerability management
- Automate compute protection
- None of these

### Best Practices marked as Not Applicable

- Provision compute from hardened images

  Out of Scope

- Reduce manual management and interactive access

  Out of Scope

- Validate software integrity

  Out of Scope

### Notes

SEC_6: Provide answers for the following:
1) Is there a process to scan software and patch vulnerabilities on a scheduled basis (Yes/No)?:

### Improvement plan

- Perform vulnerability management
- Automate compute protection

| 6. How do you protect your compute resources? |
| --- |
| Ask an expert |

  

## 7. How do you classify your data?

High risk

*This question has best practices marked as not applicable by the reviewer

### Selected choice(s)

-

### Not selected choice(s)

- Understand your data classification scheme
- Apply data protection controls based on data sensitivity
- None of these

### Best Practices marked as Not Applicable

- Automate identification and classification

  Out of Scope

- Define scalable data lifecycle management

  Out of Scope

### Notes

SEC_7: Provide answers for the following:
1) Is there a process to perform reviews to classify your data (Yes/No)?:

---

### Improvement plan

- Understand your data classification scheme
- Apply data protection controls based on data sensitivity

Ask an expert

## 8. How do you protect your data at rest?

High risk

*This question has best practices marked as not applicable by the reviewer

### Selected choice(s)

-

### Not selected choice(s)

- Enforce encryption at rest
- None of these

### Best Practices marked as Not Applicable

- Enforce access control

  Out of Scope

- Automate data at rest protection

  Out of Scope

- Implement secure key management

  Out of Scope

### Notes

SEC_8

### Improvement plan

- Enforce encryption at rest

Ask an expert

## 9. How do you protect your data in transit?

Unanswered

**Selected choice(s)**

-

**Not selected choice(s)**

- Implement secure key and certificate management
- Enforce encryption in transit
- Authenticate network communications
- None of these

**Best Practices marked as Not Applicable**

-

**Notes**

SEC_9

**Improvement plan**

Answer the question to view the improvement plan.

## 10. How do you anticipate, respond to, and recover from incidents?

High risk

*This question has best practices marked as not applicable by the reviewer

**Selected choice(s)**

-

**Not selected choice(s)**

- Identify key personnel and external resources
- Develop incident management plans
- None of these

**Best Practices marked as Not Applicable**

- Establish a framework for learning from incidents

  Out of Scope

- Develop and test security incident response playbooks

  Out of Scope

- Pre-deploy tools

  Out of Scope

- Pre-provision access

  Out of Scope

- Prepare forensic capabilities

  Out of Scope

- Run simulations

  Out of Scope

**Notes**

SEC_10: Provide answers for the following:

| 10. How do you anticipate, respond to, and recover from incidents? |
| --- |
| 1) Is there a documented process to respond, mitigate, and recover from security incidents (Yes/No)?:<br>2) Are AWS alternate contacts configured (Yes/No)?:<br><br>**Improvement plan**<br>• Identify key personnel and external resources<br>• Develop incident management plans<br><br>Ask an expert |

## 11. How do you incorporate and validate the security properties of applications throughout the design, development, and deployment lifecycle?

No improvements identified

*This question has best practices marked as not applicable by the reviewer

**Selected choice(s)**

-

**Not selected choice(s)**

- None of these

**Best Practices marked as Not Applicable**

- Automate testing throughout the development and release lifecycle
  Out of Scope
- Build a program that embeds security ownership in workload teams
  Out of Scope
- Centralize services for packages and dependencies
  Out of Scope
- Deploy software programmatically
  Out of Scope
- Conduct code reviews
  Out of Scope
- Perform regular penetration testing
  Out of Scope
- Regularly assess security properties of the pipelines
  Out of Scope

11. How do you incorporate and validate the security properties of applications throughout the design, development, and deployment lifecycle?

- Train for application security

  Out of Scope

**Notes**

SEC_11

**Improvement plan**

No risk detected for this question. No action needed.

# Reliability

### Questions answered

13/13

### Question status

- High risk: 3
- Medium risk: 2
- No improvements identified: 8
- Not Applicable: 0
- Unanswered: 0

### Pillar notes

-

## 1. How do you manage service quotas and constraints?

High risk

*This question has best practices marked as not applicable by the reviewer

### Selected choice(s)

-

### Not selected choice(s)

- Aware of service quotas and constraints
- None of these

### Best Practices marked as Not Applicable

- Automate quota management

  Out of Scope

- Accommodate fixed service quotas and constraints through architecture

  Out of Scope

- Manage service quotas across accounts and Regions

  Out of Scope

- Monitor and manage quotas

  Out of Scope

- Ensure that a sufficient gap exists between the current quotas and the maximum usage to accommodate failover

  Out of Scope

### Notes

REL_1

---

### Improvement plan

| 1. How do you manage service quotas and constraints? |
| --- |
| • Aware of service quotas and constraints<br><br>Ask an expert |

## 2. How do you plan your network topology?

No improvements identified

*This question has best practices marked as not applicable by the reviewer

**Selected choice(s)**

-

**Not selected choice(s)**

- None of these

**Best Practices marked as Not Applicable**

- Provision redundant connectivity between private networks in the cloud and on-premises environments

  Out of Scope

- Use highly available network connectivity for your workload public endpoints

  Out of Scope

- Ensure IP subnet allocation accounts for expansion and availability

  Out of Scope

- Enforce non-overlapping private IP address ranges in all private address spaces where they are connected

  Out of Scope

- Prefer hub-and-spoke topologies over many-to-many mesh

  Out of Scope

**Notes**

REL_2

| 2. How do you plan your network topology? |
| --- |
| **Improvement plan**<br>No risk detected for this question. No action needed. |

## 3. How do you design your workload service architecture?

High risk

*This question has best practices marked as not applicable by the reviewer

### Selected choice(s)

-

### Not selected choice(s)

- Build services focused on specific business domains and functionality
- Provide service contracts per API
- None of these

### Best Practices marked as Not Applicable

- Choose how to segment your workload

  Out of Scope

### Notes

REL_3: Provide answers for the following:
1) Does the solution have SLAs defined for availability and uptime (Yes/No)?:
2) Does your architecture support the SLA (Yes/No)?:

---

### Improvement plan

- Build services focused on specific business domains and functionality
- Provide service contracts per API

Ask an expert

## 4. How do you design interactions in a distributed system to prevent failures?

No improvements identified

*This question has best practices marked as not applicable by the reviewer

### Selected choice(s)

-

### Not selected choice(s)

- None of these

### Best Practices marked as Not Applicable

- Do constant work

  Out of Scope

- Make mutating operations idempotent

  Out of Scope

- Identify the kind of distributed systems you depend on

  Out of Scope

- Implement loosely coupled dependencies

  Out of Scope

### Notes

REL_4

### Improvement plan

No risk detected for this question. No action needed.

## 5. How do you design interactions in a distributed system to mitigate or withstand failures?

No improvements identified

*This question has best practices marked as not applicable by the reviewer

**Selected choice(s)**

-

**Not selected choice(s)**

- None of these

**Best Practices marked as Not Applicable**

- Set client timeouts

  Out of Scope

- Implement emergency levers

  Out of Scope

- Fail fast and limit queues

  Out of Scope

- Implement graceful degradation to transform applicable hard dependencies into soft dependencies

  Out of Scope

- Control and limit retry calls

  Out of Scope

- Make systems stateless where possible

  Out of Scope

- Throttle requests

  Out of Scope

### 5. How do you design interactions in a distributed system to mitigate or withstand failures?

**Notes**

REL_5

**Improvement plan**

No risk detected for this question. No action needed.

## 6. How do you monitor workload resources?

No improvements identified

*This question has best practices marked as not applicable by the reviewer

### Selected choice(s)

-

### Not selected choice(s)

- None of these

### Best Practices marked as Not Applicable

- Automate responses (Real-time processing and alarming)
  Out of Scope
- Monitor end-to-end tracing of requests through your system
  Out of Scope
- Monitor all components for the workload (Generation)
  Out of Scope
- Define and calculate metrics (Aggregation)
  Out of Scope
- Send notifications (Real-time processing and alarming)
  Out of Scope
- Regularly review monitoring scope and metrics
  Out of Scope
- Analyze logs
  Out of Scope

### Notes

REL_6

  

## 6. How do you monitor workload resources?

### Improvement plan

No risk detected for this question. No action needed.

## 7. How do you design your workload to adapt to changes in demand?

No improvements identified

*This question has best practices marked as not applicable by the reviewer

**Selected choice(s)**

-

**Not selected choice(s)**

- None of these

**Best Practices marked as Not Applicable**

- Use automation when obtaining or scaling resources

  Out of Scope

- Load test your workload

  Out of Scope

- Obtain resources upon detection that more resources are needed for a workload

  Out of Scope

- Obtain resources upon detection of impairment to a workload

  Out of Scope

**Notes**

REL_7

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**Improvement plan**

No risk detected for this question. No action needed.

## 8. How do you implement change?

No improvements identified

*This question has best practices marked as not applicable by the reviewer

**Selected choice(s)**

-

**Not selected choice(s)**

- None of these

**Best Practices marked as Not Applicable**

- Deploy changes with automation
  Out of Scope
- Integrate functional testing as part of your deployment
  Out of Scope
- Deploy using immutable infrastructure
  Out of Scope
- Use runbooks for standard activities such as deployment
  Out of Scope
- Integrate resiliency testing as part of your deployment
  Out of Scope

**Notes**

REL_8

---

**Improvement plan**

No risk detected for this question. No action needed.

  

## 9. How do you back up data?

Medium risk

*This question has best practices marked as not applicable by the reviewer

### Selected choice(s)

-

### Not selected choice(s)

- Perform data backup automatically
- Perform periodic recovery of the data to verify backup integrity and processes
- None of these

### Best Practices marked as Not Applicable

- Identify and back up all data that needs to be backed up, or reproduce the data from sources

  Out of Scope

- Secure and encrypt backups

  Out of Scope

### Notes

REL_9: Provide answers for the following:
1) Is all critical production data backed up automatically (Yes/No)?:
2) Do you perform periodic recovery of the data to verify backup integrity and processes (Yes/No)?:
3) Provide the date for the last time your disaster recovery plan was tested:
4) Do you test data recovery after making significant changes to the solution (Yes/No)?:

| 9. How do you back up data? |
|---|
| **Improvement plan**<br>• Perform data backup automatically<br>• Perform periodic recovery of the data to verify backup integrity and processes<br><br>Ask an expert |

  

## 10. How do you use fault isolation to protect your workload?

No improvements identified

*This question has best practices marked as not applicable by the reviewer

### Selected choice(s)

-

### Not selected choice(s)

- None of these

### Best Practices marked as Not Applicable

- Deploy the workload to multiple locations

  Out of Scope

- Automate recovery for components constrained to a single location

  Out of Scope

- Use bulkhead architectures to limit scope of impact

  Out of Scope

### Notes

REL_10

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### Improvement plan

No risk detected for this question. No action needed.

## 11. How do you design your workload to withstand component failures?

No improvements identified

*This question has best practices marked as not applicable by the reviewer

**Selected choice(s)**

-

**Not selected choice(s)**

- None of these

**Best Practices marked as Not Applicable**

- Automate healing on all layers

  Out of Scope
- Rely on the data plane and not the control plane during recovery

  Out of Scope
- Fail over to healthy resources

  Out of Scope
- Monitor all components of the workload to detect failures

  Out of Scope
- Send notifications when events impact availability

  Out of Scope
- Architect your product to meet availability targets and uptime service level agreements (SLAs)

  Out of Scope
- Use static stability to prevent bimodal behavior

  Out of Scope

  

| 11. How do you design your workload to withstand component failures? |
| --- |
| **Notes**<br>REL_11<br><br>**Improvement plan**<br>No risk detected for this question. No action needed. |

## 12. How do you test reliability?

⚠ Medium risk

*This question has best practices marked as not applicable by the reviewer

### Selected choice(s)

-

### Not selected choice(s)

- Test resiliency using chaos engineering
- Conduct game days regularly
- None of these

### Best Practices marked as Not Applicable

- Use playbooks to investigate failures

  Out of Scope

- Perform post-incident analysis

  Out of Scope

- Test scalability and performance requirements

  Out of Scope

### Notes

REL_12: Provide answers for the following:
1) Is resiliency testing conducted for the solution (Yes/No)?:
2) What was the date of the last resiliency testing?:

### Improvement plan

- Test resiliency using chaos engineering
- Conduct game days regularly

  

| 12. How do you test reliability? |
| --- |
| Ask an expert |

## 13. How do you plan for disaster recovery (DR)?

High risk

*This question has best practices marked as not applicable by the reviewer

### Selected choice(s)

-

### Not selected choice(s)

- Define recovery objectives for downtime and data loss
- Test disaster recovery implementation to validate the implementation
- None of these

### Best Practices marked as Not Applicable

- Automate recovery

  Out of Scope

- Manage configuration drift at the DR site or Region

  Out of Scope

- Use defined recovery strategies to meet the recovery objectives

  Out of Scope

### Notes

REL_13: Provide answers for the following:
1) Is RPO defined for the solution (Yes/No)?:
2) What is your RPO?:
3) Is RTO defined for the solution (Yes/No)?:
4) What is your RTO?:

---

### Improvement plan

- Define recovery objectives for downtime and data loss

| 13. How do you plan for disaster recovery (DR)? |
| --- |
| • Test disaster recovery implementation to validate the implementation<br><br>Ask an expert |

# Performance Efficiency

### Questions answered

5/5

### Question status

- High risk: 0
- Medium risk: 0
- No improvements identified: 5
- Not Applicable: 0
- Unanswered: 0

### Pillar notes

-

## 1. How do you select the appropriate cloud resources and architecture patterns for your workload?

No improvements identified

*This question has best practices marked as not applicable by the reviewer

**Selected choice(s)**

-

**Not selected choice(s)**

- None of these

**Best Practices marked as Not Applicable**

- Evaluate how trade-offs impact customers and architecture efficiency
  Out of Scope
- Factor cost into architectural decisions
  Out of Scope
- Use guidance from your cloud provider or an appropriate partner to learn about architecture patterns and best practices
  Out of Scope
- Learn about and understand available cloud services and features
  Out of Scope
- Use benchmarking to drive architectural decisions
  Out of Scope
- Use a data-driven approach for architectural choices
  Out of Scope
- Use policies and reference architectures
  Out of Scope

| 1. How do you select the appropriate cloud resources and architecture patterns for your workload? |
| --- |
| **Notes**<br>PERF_1<br><br>**Improvement plan**<br>No risk detected for this question. No action needed. |

## 2. How do you select and use compute resources in your workload?

No improvements identified

*This question has best practices marked as not applicable by the reviewer

**Selected choice(s)**

-

**Not selected choice(s)**

- None of these

**Best Practices marked as Not Applicable**

- Collect compute-related metrics
  Out of Scope
- Use optimized hardware-based compute accelerators
  Out of Scope
- Configure and right-size compute resources
  Out of Scope
- Scale your compute resources dynamically
  Out of Scope
- Select the best compute options for your workload
  Out of Scope
- Understand the available compute configuration and features
  Out of Scope

**Notes**

PERF_2

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**Improvement plan**

| 2. How do you select and use compute resources in your workload? |
| --- |
| No risk detected for this question. No action needed. |

  

## 3. How do you store, manage, and access data in your workload?

No improvements identified

*This question has best practices marked as not applicable by the reviewer

### Selected choice(s)

-

### Not selected choice(s)

- None of these

### Best Practices marked as Not Applicable

- Implement data access patterns that utilize caching

  Out of Scope
- Collect and record data store performance metrics

  Out of Scope
- Evaluate available configuration options for data store

  Out of Scope
- Implement strategies to improve query performance in data store

  Out of Scope
- Use purpose-built data store that best support your data access and storage requirements

  Out of Scope

### Notes

PERF_3

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### Improvement plan

No risk detected for this question. No action needed.

## 4. How do you select and configure networking resources in your workload?

No improvements identified

*This question has best practices marked as not applicable by the reviewer

**Selected choice(s)**

-

**Not selected choice(s)**

- None of these

**Best Practices marked as Not Applicable**

- Choose appropriate dedicated connectivity or VPN for your workload
  Out of Scope
- Choose network protocols to improve performance
  Out of Scope
- Choose your workload's location based on network requirements
  Out of Scope
- Evaluate available networking features
  Out of Scope
- Use load balancing to distribute traffic across multiple resources
  Out of Scope
- Optimize network configuration based on metrics
  Out of Scope
- Understand how networking impacts performance
  Out of Scope

**Notes**

| 4. How do you select and configure networking resources in your workload? |
| --- |
| PERF_4<br><br>**Improvement plan**<br>No risk detected for this question. No action needed. |

## 5. What process do you use to support more performance efficiency for your workload?

No improvements identified

*This question has best practices marked as not applicable by the reviewer

**Selected choice(s)**

-

**Not selected choice(s)**

- None of these

**Best Practices marked as Not Applicable**

- Use automation to proactively remediate performance-related issues
  Out of Scope
- Establish key performance indicators (KPIs) to measure workload health and performance
  Out of Scope
- Keep your workload and services up-to-date
  Out of Scope
- Load test your workload
  Out of Scope
- Review metrics at regular intervals
  Out of Scope
- Use monitoring solutions to understand the areas where performance is most critical
  Out of Scope
- Define a process to improve workload performance

5. What process do you use to support more performance efficiency for your workload?

Out of Scope

**Notes**

PERF_5

**Improvement plan**

No risk detected for this question. No action needed.

  

# Cost Optimization

### Questions answered

11/11

### Question status

- High risk: 0
- Medium risk: 0
- No improvements identified: 11
- Not Applicable: 0
- Unanswered: 0

### Pillar notes

-

## 1. How do you implement cloud financial management?

No improvements identified

*This question has best practices marked as not applicable by the reviewer

**Selected choice(s)**

-

**Not selected choice(s)**

- None of these

**Best Practices marked as Not Applicable**

- Establish cloud budgets and forecasts
  Out of Scope
- Implement cost awareness in your organizational processes
  Out of Scope
- Create a cost-aware culture
  Out of Scope
- Establish ownership of cost optimization
  Out of Scope
- Establish a partnership between finance and technology
  Out of Scope
- Monitor cost proactively
  Out of Scope
- Quantify business value from cost optimization
  Out of Scope
- Keep up-to-date with new service releases
  Out of Scope

  

## 1. How do you implement cloud financial management?

- Report and notify on cost optimization

  Out of Scope

**Notes**

COST_1

---

**Improvement plan**

No risk detected for this question. No action needed.

## 2. How do you govern usage?

No improvements identified

*This question has best practices marked as not applicable by the reviewer

### Selected choice(s)

-

### Not selected choice(s)

- None of these

### Best Practices marked as Not Applicable

- Implement an account structure

  Out of Scope

- Implement cost controls

  Out of Scope

- Implement goals and targets

  Out of Scope

- Implement groups and role

  Out of Scope

- Develop policies based on your organization requirements

  Out of Scope

- Track project lifecycle

  Out of Scope

### Notes

COST_2

### Improvement plan

| 2. How do you govern usage? |
| --- |
| No risk detected for this question. No action needed. |

## 3. How do you monitor your cost and usage?

No improvements identified

*This question has best practices marked as not applicable by the reviewer

### Selected choice(s)

-

### Not selected choice(s)

- None of these

### Best Practices marked as Not Applicable

- Allocate costs based on workload metrics

  Out of Scope
- Configure billing and cost management tools

  Out of Scope
- Identify cost attribution categories

  Out of Scope
- Establish organization metrics

  Out of Scope
- Configure detailed information sources

  Out of Scope
- Add organization information to cost and usage

  Out of Scope

### Notes

COST_3

### Improvement plan

| 3. How do you monitor your cost and usage? |
| --- |
| No risk detected for this question. No action needed. |

## 4. How do you decommission resources?

No improvements identified

*This question has best practices marked as not applicable by the reviewer

### Selected choice(s)

-

### Not selected choice(s)

- None of these

### Best Practices marked as Not Applicable

- Enforce data retention policies

  Out of Scope

- Decommission resources automatically

  Out of Scope

- Decommission resources

  Out of Scope

- Implement a decommissioning process

  Out of Scope

- Track resources over their life time

  Out of Scope

### Notes

COST_4

### Improvement plan

No risk detected for this question. No action needed.

## 5. How do you evaluate cost when you select services?

No improvements identified

*This question has best practices marked as not applicable by the reviewer

### Selected choice(s)

-

### Not selected choice(s)

- None of these

### Best Practices marked as Not Applicable

- Analyze all components of this workload

  Out of Scope

- Perform cost analysis for different usage over time

  Out of Scope

- Select software with cost effective licensing

  Out of Scope

- Identify organization requirements for cost

  Out of Scope

- Select components of this workload to optimize cost in line with organization priorities

  Out of Scope

- Perform a thorough analysis of each component

  Out of Scope

### Notes

COST_5

| 5. How do you evaluate cost when you select services? |
| --- |
| **Improvement plan**<br>No risk detected for this question. No action needed. |

## 6. How do you meet cost targets when you select resource type, size and number?

No improvements identified

*This question has best practices marked as not applicable by the reviewer

**Selected choice(s)**

-

**Not selected choice(s)**

- None of these

**Best Practices marked as Not Applicable**

- Perform cost modeling

  Out of Scope

- Select resource type, size, and number based on data

  Out of Scope

- Select resource type, size, and number automatically based on metrics

  Out of Scope

- Consider using shared resources

  Out of Scope

**Notes**

COST_6

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**Improvement plan**

No risk detected for this question. No action needed.

## 7. How do you use pricing models to reduce cost?

No improvements identified

*This question has best practices marked as not applicable by the reviewer

### Selected choice(s)

-

### Not selected choice(s)

- None of these

### Best Practices marked as Not Applicable

- Perform pricing model analysis

  Out of Scope

- Implement pricing models for all components of this workload

  Out of Scope

- Perform pricing model analysis at the management account level

  Out of Scope

- Choose Regions based on cost

  Out of Scope

- Select third-party agreements with cost-efficient terms

  Out of Scope

### Notes

COST_7

### Improvement plan

No risk detected for this question. No action needed.

  

## 8. How do you plan for data transfer charges?

No improvements identified

*This question has best practices marked as not applicable by the reviewer

### Selected choice(s)

-

### Not selected choice(s)

- None of these

### Best Practices marked as Not Applicable

- Implement services to reduce data transfer costs

  Out of Scope

- Perform data transfer modeling

  Out of Scope

- Select components to optimize data transfer cost

  Out of Scope

### Notes

COST_8

### Improvement plan

No risk detected for this question. No action needed.

## 9. How do you manage demand, and supply resources?

No improvements identified

*This question has best practices marked as not applicable by the reviewer

**Selected choice(s)**

-

**Not selected choice(s)**

- None of these

**Best Practices marked as Not Applicable**

- Implement a buffer or throttle to manage demand

  Out of Scope

- Perform an analysis on the workload demand

  Out of Scope

- Supply resources dynamically

  Out of Scope

**Notes**

COST_9

**Improvement plan**

No risk detected for this question. No action needed.

## 10. How do you evaluate new services?

No improvements identified

*This question has best practices marked as not applicable by the reviewer

### Selected choice(s)

-

### Not selected choice(s)

- None of these

### Best Practices marked as Not Applicable

- Develop a workload review process

  Out of Scope

- Review and analyze this workload regularly

  Out of Scope

### Notes

COST_10

### Improvement plan

No risk detected for this question. No action needed.

  

## 11. How do you evaluate the cost of effort?

No improvements identified

*This question has best practices marked as not applicable by the reviewer

**Selected choice(s)**

-

**Not selected choice(s)**

- None of these

**Best Practices marked as Not Applicable**

- Perform automation for operations

  Out of Scope

**Notes**

COST_11

**Improvement plan**

No risk detected for this question. No action needed.

# Sustainability

### Questions answered

6/6

### Question status

- High risk: 0
- Medium risk: 0
- No improvements identified: 6
- Not Applicable: 0
- Unanswered: 0

### Pillar notes

-

## 1. How do you select Regions for your workload?

No improvements identified

*This question has best practices marked as not applicable by the reviewer

### Selected choice(s)

-

### Not selected choice(s)

- None of these

### Best Practices marked as Not Applicable

- Choose Region based on both business requirements and sustainability goals

  Out of Scope

### Notes

SUS_1

### Improvement plan

No risk detected for this question. No action needed.

## 2. How do you align cloud resources to your demand?

No improvements identified

*This question has best practices marked as not applicable by the reviewer

### Selected choice(s)

-

### Not selected choice(s)

- None of these

### Best Practices marked as Not Applicable

- Scale workload infrastructure dynamically

  Out of Scope
- Align SLAs with sustainability goals

  Out of Scope
- Stop the creation and maintenance of unused assets

  Out of Scope
- Optimize geographic placement of workloads based on their networking requirements

  Out of Scope
- Optimize team member resources for activities performed

  Out of Scope
- Implement buffering or throttling to flatten the demand curve

  Out of Scope

### Notes

SUS_2

  

| 2. How do you align cloud resources to your demand? |
| --- |
| **Improvement plan**<br>No risk detected for this question. No action needed. |

  

3. How do you take advantage of software and architecture patterns to support your sustainability goals?

No improvements identified

*This question has best practices marked as not applicable by the reviewer

**Selected choice(s)**

-

**Not selected choice(s)**

- None of these

**Best Practices marked as Not Applicable**

- Optimize software and architecture for asynchronous and scheduled jobs
  Out of Scope
- Remove or refactor workload components with low or no use
  Out of Scope
- Optimize areas of code that consume the most time or resources
  Out of Scope
- Optimize impact on devices and equipment
  Out of Scope
- Use software patterns and architectures that best support data access and storage patterns
  Out of Scope

**Notes**

SUS_3

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**Improvement plan**

| 3. How do you take advantage of software and architecture patterns to support your sustainability goals? |
| --- |
| No risk detected for this question. No action needed. |

4. How do you take advantage of data management policies and patterns to support your sustainability goals?

No improvements identified

*This question has best practices marked as not applicable by the reviewer

**Selected choice(s)**

-

**Not selected choice(s)**

- None of these

**Best Practices marked as Not Applicable**

- Implement a data classification policy
  Out of Scope
- Use technologies that support data access and storage patterns
  Out of Scope
- Use policies to manage the lifecycle of your datasets
  Out of Scope
- Use elasticity and automation to expand block storage or file system
  Out of Scope
- Remove unneeded or redundant data
  Out of Scope
- Use shared file systems or storage to access common data
  Out of Scope
- Minimize data movement across networks
  Out of Scope
- Back up data only when difficult to recreate

4. How do you take advantage of data management policies and patterns to support your sustainability goals?

Out of Scope

**Notes**

SUS_4

**Improvement plan**

No risk detected for this question. No action needed.

  

| 5. How do you select and use cloud hardware and services in your architecture to support your sustainability goals? |
|---|
| No improvements identified<br>*This question has best practices marked as not applicable by the reviewer<br><br>**Selected choice(s)**<br>-<br><br>**Not selected choice(s)**<br>• None of these<br><br>**Best Practices marked as Not Applicable**<br>• Use the minimum amount of hardware to meet your needs<br>Out of Scope<br>• Use instance types with the least impact<br>Out of Scope<br>• Use managed services<br>Out of Scope<br>• Optimize your use of hardware-based compute accelerators<br>Out of Scope<br><br>**Notes**<br>SUS_5<br><br>**Improvement plan**<br>No risk detected for this question. No action needed. |

## 6. How do your organizational processes support your sustainability goals?

No improvements identified

*This question has best practices marked as not applicable by the reviewer

**Selected choice(s)**

-

**Not selected choice(s)**

- None of these

**Best Practices marked as Not Applicable**

- Communicate and cascade your sustainability goals
  Out of Scope
- Adopt methods that can rapidly introduce sustainability improvements
  Out of Scope
- Keep your workload up-to-date
  Out of Scope
- Increase utilization of build environments
  Out of Scope
- Use managed device farms for testing
  Out of Scope

**Notes**

SUS_6

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**Improvement plan**

No risk detected for this question. No action needed.

  